

ECEN-629: Homework 3

Due on October 21, 2025 at 12:45 pm

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Problem 1

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, and define

$$g(x, t) = f(x/t), \quad \text{dom } g = \{(x, t) \mid x/t \in \text{dom } f, t > 0\}.$$

Show that g is quasiconvex, and then write down the optimization procedure similarly to that on page 17 of the slides of Chapter 4, specialized to this problem.

Hint: consider the function $t \cdot f(x/t)$.

Solution

From Chapter 3, we know that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasiconvex if $\text{dom } f$ is convex and the sublevel sets

$$S_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$$

are convex for all α . For g :

$$S_\alpha = \{(x, t) \in \text{dom } g \mid g(x, t) \leq \alpha\} = \{(x, t) \mid f(x/t) \leq \alpha, t > 0\}.$$

Since $t > 0$, the sublevel sets of g are scaled sublevel sets of f , which are convex. Thus, g is quasiconvex.

Algorithm 1 Quasiconvex Optimization

Require: $l \leq p^*$, $u \geq p^*$, tolerance $\varepsilon > 0$

1: **repeat**

2: $t \leftarrow (l + u)/2$

3: Solve the convex feasibility problem:

$$f(x/t) \leq \alpha, \quad t > 0$$

4: **if** feasible **then**

5: $u \leftarrow t$

6: **else**

7: $l \leftarrow t$

8: **until** $u - l \leq \varepsilon$

9: **return** $t^* \approx u$

Problem 2

Some non-convex problems can also be easy to solve. Find the optimal solution for the following problems in closed form.

(a) For $x \in \mathbb{R}^n$

$$\begin{aligned} \text{minimize: } & \|x\|_2 \\ \text{subject to: } & 10 \geq \|x\|_1 \geq 1. \end{aligned}$$

(b) Let $x = (x_1, x_2) \in \mathbb{R}^2$

$$\begin{aligned} \text{minimize: } & x_1 + x_2 \\ \text{subject to: } & x_1 + 3x_2 = 5; \\ & (x_1 - 3)^2 + (x_2 - 4)^2 \geq 25. \end{aligned}$$

Solution

Part (a): The objective function $f_0(x) = \sqrt{\sum_{i=1}^n |x_i|^2}$. Choosing the lower bound of the inequality constraint requires $\|x\|_1 = 1$. The ℓ_2 norm is minimal when each component of $x_i \leq 1$. Thus we aim for each component to be minimal, but sum to 1. Thus, $x_i^* = 1/n$ for $i = 1, 2, \dots, n$.

Part (b): This problem is unbounded below, so the optimal solution is $-\infty$.

Problem 3

Huber's penalty function is defined as

$$\phi(u) = \begin{cases} u^2, & |u| \leq M, \\ M(2|u| - M), & |u| > M. \end{cases}$$

- (a) Is the function $\phi(u)$ convex in general?
 (b) Show that the minimization problem

$$\text{minimize: } \sum_{i=1}^n \phi(a_i^T x - b_i)$$

is equivalent to the following problem

$$\begin{aligned} \text{minimize: } & \sum_{i=1}^n (u_i^2 + 2Mv_i) \\ \text{subject to: } & -u - v \preceq Ax - b \preceq u + v \\ & 0 \preceq u \preceq M\mathbf{1} \\ & v \succeq 0. \end{aligned}$$

- (c) A general form of Huber's penalty function with parameters (α, β, γ) is

$$\phi(u) = \begin{cases} \alpha \cdot u^2, & |u| \leq M, \\ \beta M(\gamma|u| - M), & |u| > M. \end{cases}$$

For any fixed value of $\alpha > 0$, find the valid assignment of (β, γ) in terms of α such that the penalty function is convex.

Solution

Part (a): Yes, $\phi(u)$ is convex. Quadratic functions are convex, and $M(2|u| - M)$ is also convex.

Part (b): Here, expanding the objective function gives

$$\phi(a_i^T x - b_i) = \begin{cases} (a_i^T x - b_i)^2, & |a_i^T x - b_i| \leq M, \\ M(2|a_i^T x - b_i| - M), & |a_i^T x - b_i| > M. \end{cases}$$

where the constraints become

$$-u - v \preceq Ax - b \preceq u + v, \quad 0 \preceq u \preceq M\mathbf{1}, \quad v \succeq 0.$$

Part (c): By continuity, we require

$$\alpha \cdot M^2 = \beta M(\gamma M - M) \implies \beta = \frac{\alpha M^2}{M(\gamma M - M)} = \frac{\alpha}{\gamma - 1}.$$

Problem 4

Write the following problems as equivalent LPs:

(a)

$$\begin{aligned} &\text{minimize:} && \|x\|_\infty \\ &\text{subject to:} && \|Ax - b\|_1 \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(b)

$$\begin{aligned} &\text{minimize:} && \|x\|_\infty \\ &\text{subject to:} && \|Ax - b\|_\infty \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(c)

$$\begin{aligned} &\text{minimize:} && \|x\|_1 \\ &\text{subject to:} && \|Ax - b\|_\infty \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(d)

$$\begin{aligned} &\text{minimize:} && \|x\|_1 \\ &\text{subject to:} && \|Ax - b\|_1 \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

For the problems above, what if we exchange the objective function and the constraint, can you still write down the problems as LPs?

Solution

Part (a):

$$\begin{aligned} &\text{minimize:} && t \\ &\text{subject to:} && \mathbf{1} \cdot -t \preceq x \preceq \mathbf{1}t \\ &&& -\mathbf{1}a \preceq Ax - b \preceq \mathbf{1}a \\ &&& \mathbf{1}^T a \preceq 1, \quad a \succeq 0. \end{aligned}$$

Part (b):

$$\begin{array}{ll}\text{minimize:} & t \\ \text{subject to:} & \mathbf{1} \cdot -t \preceq x \preceq \mathbf{1}t \\ & -\mathbf{1} \preceq Ax - b \preceq \mathbf{1}.\end{array}$$

Part (c):

$$\begin{array}{ll}\text{minimize:} & \mathbf{1}t \\ \text{subject to:} & -t \preceq x \preceq t \\ & -\mathbf{1} \preceq Ax - b \preceq \mathbf{1}.\end{array}$$

Part (d):

$$\begin{array}{ll}\text{minimize:} & \mathbf{1}t \\ \text{subject to:} & -t \preceq x \preceq t \\ & -\mathbf{1}a \preceq Ax - b \preceq \mathbf{1}a \\ & \mathbf{1}^T a \preceq 1, \quad a \succeq 0.\end{array}$$

Problem 5

Consider the optimization problem in $\mathbb{R}^2$

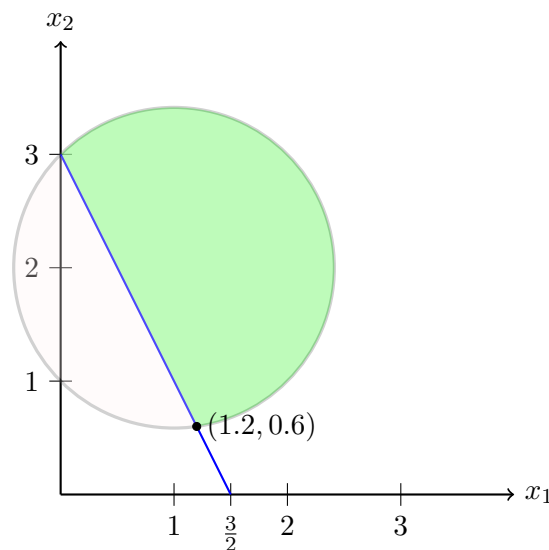
$$\begin{aligned} \text{minimize: } & f_0(x_1, x_2) \\ \text{subject to: } & (x_1 - 1)^2 + (x_2 - 2)^2 \leq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0, x_2 \geq 0. \end{aligned}$$

First make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value, using two methods: geometrically and the first-order optimality condition. Note that you should not use KKT directly here, but the first-order condition in Chapter 4.

(a) $f_0(x_1, x_2) = x_1 + x_2$;

(b) $f_0(x_1, x_2) = x_1^2 + x_2^2$.

Solution



Part (a): Geometrically, the optimal solution lies at the lower intersection, which is $x^* = (1.2, 0.6)$, $f(x^*) = 1.2 + 0.6 = 1.8$.

Part (b): Geometrically, the optimal solution lies at the point closest to the origin, with preference for values less than 1. In this case that point is also $x^* = (1.2, 0.6)$, $f(x^*) = 1.2^2 + 0.6^2 = 1.8$.

Problem 6

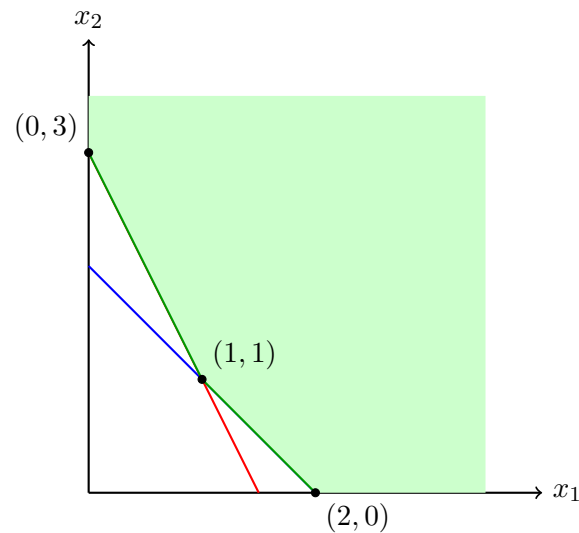
Consider the optimization problem

$$\begin{aligned} \text{minimize: } & f_0(x_1, x_2) \\ \text{subject to: } & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0, x_2 \geq 0. \end{aligned}$$

First make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value, using two methods: geometrically and the first-order optimality condition.

- (a) $f_0(x_1, x_2) = 3x_1 + 2x_2$;
- (b) $f_0(x_1, x_2) = \max\{x_1, x_2\}$.

Solution



Part (a): Geometrically, it is clear that the optimal solution is $x = (1, 1)$, $f(x) = 3(1) + 2(1) = 5$.

Part (b): Geometrically, it is clear that the optimal solution is also $x = (1, 1)$, $f(x) = \max\{1, 1\} = 1$.

Problem 7

Give explicit solutions for the following LP problems over $\mathbb{R}^n$

(a)

$$\begin{aligned} \text{minimize: } & c^T x \\ \text{subject to: } & a^T x \leq b, \end{aligned}$$

where $a \neq 0$.

(b)

$$\begin{aligned} \text{minimize: } & c^T x \\ \text{subject to: } & \mathbf{1}^T x \leq \alpha \\ & 0 \cdot \mathbf{1} \preceq l \preceq x \preceq u. \end{aligned}$$

Solution

Part (a): Because of the half-space constraint, this problem is unbounded below, so the optimal solution is $-\infty$.

Part (b): Splitting by components gives $c_i x_i$ subject to $0 \leq l_i \leq x_i \leq u_i$ for each $i = 1, 2, \dots, n$.

$$x_i^* = \begin{cases} l_i, & c_i > 0, \\ u_i, & c_i < 0, \end{cases}$$

where $\sum x_i^* \leq \alpha$.

Problem 8

Find the dual optimization problems for the following optimization problems

(a)

$$\begin{aligned} \text{minimize: } & c^T x \\ \text{subject to: } & Ax = b \\ & l \preceq x \preceq u, \end{aligned}$$

where $x \in \mathbb{R}^n$ is the variable, and $l, u \in \mathbb{R}^n$

(b) Recall last time you wrote the following problem as an equivalent LP:

$$\begin{aligned} \text{minimize: } & \|x\|_\infty \\ \text{subject to: } & \|Ax - b\|_\infty \leq 1. \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Find the dual problem for the equivalent LP.

(c)

$$\begin{aligned} \text{minimize: } & \sum_{i=1}^n x_i \log x_i \\ \text{subject to: } & Ax \succeq b, \\ & c^T x = d, \end{aligned}$$

whose domain is $\mathbb{R}_+^n$, i.e., $x_i > 0$ for all $i = 1, 2, \dots, n$. Assume $c \succeq 0$.

(d)

$$\begin{aligned} \text{minimize: } & x^T Q x \\ \text{subject to: } & Ax = b, \end{aligned}$$

where Q is a positive definite matrix.

Solution

Part (a):

$$\begin{aligned} g(\nu, \lambda_1, \lambda_2) &= \inf_{x \in \mathcal{D}} L(x, \nu, \lambda_1, \lambda_2) \\ &= \inf_{x \in \mathcal{D}} (c^T x + \nu^T (Ax - b) + \lambda_1^T (x - u) + \lambda_2^T (l - x)) \\ &= \inf_{x \in \mathcal{D}} (\lambda_2^T l - \lambda_1^T u - \nu^T b + (c^T + \nu^T A + \lambda_1^T - \lambda_2^T)x) \\ &= \lambda_2^T l - \lambda_1^T u - \nu^T b \end{aligned}$$

where we maximize $g(\lambda, \nu_1, \nu_2)$ subject to $\lambda \succeq 0$.

Part (b):

$$\begin{aligned}
g(\lambda_1, \lambda_2, \lambda_3, \lambda_4) &= \inf_{t \in \mathcal{D}} L(t, \lambda_1, \lambda_2, \lambda_3, \lambda_4) \\
&= \inf_{t \in \mathcal{D}} (t + \lambda_1^T(x - t) + \lambda_2^T(-t - x) + \lambda_3^T(Ax - b - t) + \lambda_4^T(-t - Ax + b)) \\
&= t(1 - \lambda_1^T - \lambda_2^T - \lambda_3^T - \lambda_4^T) + b^T(\lambda_3 - \lambda_4)
\end{aligned}$$

where we maximize $g(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ subject to $\lambda \succeq 0$.

Part (c):

$$\begin{aligned}
g(\lambda, \nu) &= \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \\
&= \inf_{x \in \mathcal{D}} \left(\sum_{i=1}^n x_i \log x_i + \lambda^T(b - Ax) + \nu(c^T x - d) \right)
\end{aligned}$$

where we maximize $g(\lambda, \nu)$.

Part (d):

$$\begin{aligned}
g(\nu) &= \inf_x L(x, \nu) \\
&= \inf_x (x^T Q x + \nu^T(Ax - b)) \\
&= (-\frac{1}{2}A^T \nu)^T Q (-\frac{1}{2}A^T \nu) + \nu^T(A(-\frac{1}{2}A^T \nu) - b)
\end{aligned}$$

where we maximize $g(\nu)$.